

# Linear Programming Report

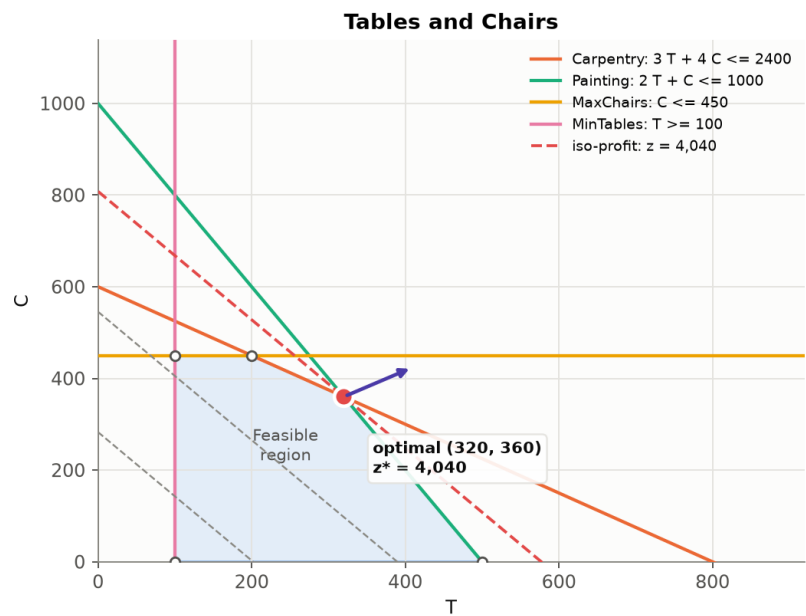
## Sensitivity Analysis & Duality

### Tables and Chairs

A shop produces tables (T) and chairs (C). Maximize profit subject to carpentry and painting hours, a market limit on chairs, and a minimum number of tables.

Source: Lecture slides 6-19 (graphical sensitivity analysis)

|                     |                                                  |
|---------------------|--------------------------------------------------|
| Problem type        | MAX, 2 variables, 4 constraints                  |
| Status              | OPTIMAL                                          |
| Optimal Profit (\$) | 4,040                                            |
| Dual status         | OPTIMAL                                          |
| Prepared by         | Dr. Islam Ali                                    |
| Course              | Sensitivity Analysis and Duality (Dr. Islam Ali) |
| Date                | 28 September 2026, 08:44                         |



# 1. Model formulation

MAX  $z = 7T + 5C$   
 subject to  
 $3T + 4C \leq 2400$  (Carpentry)  
 $2T + C \leq 1000$  (Painting)  
 $C \leq 450$  (MaxChairs)  
 $T \geq 100$  (MinTables)  
 $T \geq 0, C \geq 0$

## Data table

| Constraint  | T | C | Sense  | RHS   |
|-------------|---|---|--------|-------|
| Carpentry   | 3 | 4 | $\leq$ | 2,400 |
| Painting    | 2 | 1 | $\leq$ | 1,000 |
| MaxChairs   | 0 | 1 | $\leq$ | 450   |
| MinTables   | 1 | 0 | $\geq$ | 100   |
| Objective c | 7 | 5 | max    | —     |

## Definitions

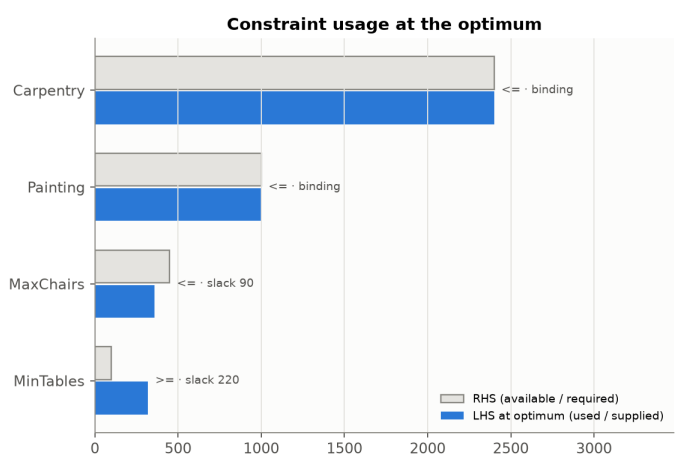
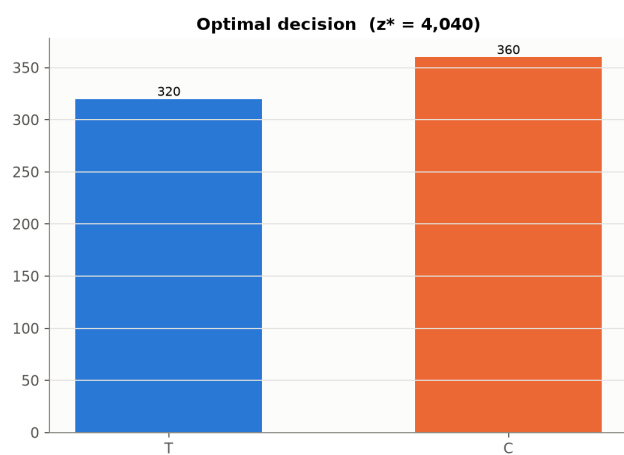
- **T**: number of tables produced
- **C**: number of chairs produced
- **Carpentry**: carpentry hours available
- **Painting**: painting hours available
- **MaxChairs**: maximum number of chairs
- **MinTables**: minimum number of tables

## 2. Optimal solution

Optimal Profit (\$): **4,040**

| Variable | Description               | Optimal value | Obj. coefficient | Contribution |
|----------|---------------------------|---------------|------------------|--------------|
| T        | number of tables produced | 320           | 7                | 2,240        |
| C        | number of chairs produced | 360           | 5                | 1,800        |

| Constraint | LHS   | Sense | RHS   | Slack/Surplus | Binding |
|------------|-------|-------|-------|---------------|---------|
| Carpentry  | 2,400 | <=    | 2,400 | 0             | yes     |
| Painting   | 1,000 | <=    | 1,000 | 0             | yes     |
| MaxChairs  | 360   | <=    | 450   | 90            | no      |
| MinTables  | 320   | >=    | 100   | 220           | no      |



### Corner points

| T   | C   | Profit (\$) | Optimal? |
|-----|-----|-------------|----------|
| 100 | 0   | 700         |          |
| 500 | 0   | 3,500       |          |
| 320 | 360 | 4,040       | ✓        |
| 200 | 450 | 3,650       |          |
| 100 | 450 | 2,950       |          |

### 3. Sensitivity analysis

#### Objective-coefficient ranging (variable cells)

| Variable | Value | Reduced cost | Obj. coef | Allow. + | Allow. – | Low  | High  |
|----------|-------|--------------|-----------|----------|----------|------|-------|
| T        | 320   | 0            | 7         | 3        | 3.25     | 3.75 | 10    |
| C        | 360   | 0            | 5         | 4.333    | 1.5      | 3.5  | 9.333 |

#### Right-hand-side ranging (constraints)

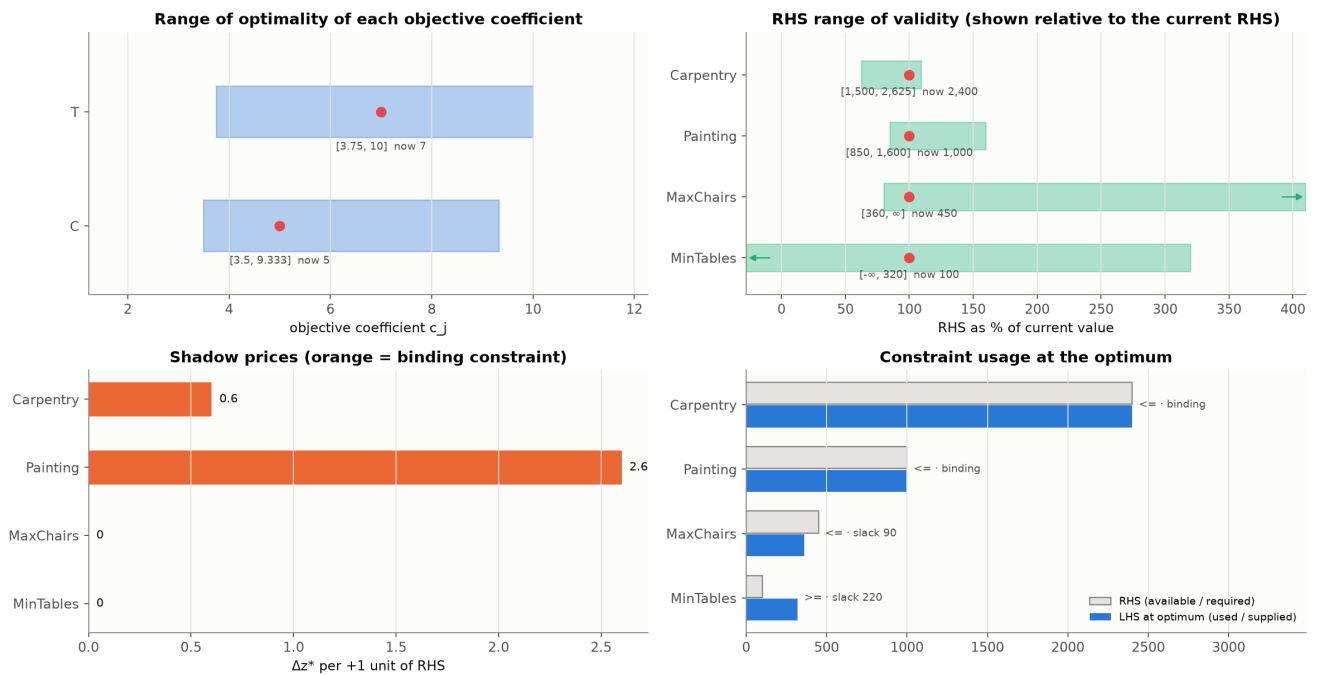
| Constraint | LHS   | Sense | RHS   | Slack | Shadow price | Allow. + | Allow. – | Low       | High     |
|------------|-------|-------|-------|-------|--------------|----------|----------|-----------|----------|
| Carpentry  | 2,400 | <=    | 2,400 | 0     | 0.6          | 225      | 900      | 1,500     | 2,625    |
| Painting   | 1,000 | <=    | 1,000 | 0     | 2.6          | 600      | 150      | 850       | 1,600    |
| MaxChairs  | 360   | <=    | 450   | 90    | 0            | $\infty$ | 90       | 360       | $\infty$ |
| MinTables  | 320   | >=    | 100   | 220   | 0            | 220      | $\infty$ | $-\infty$ | 320      |

#### Interpretation

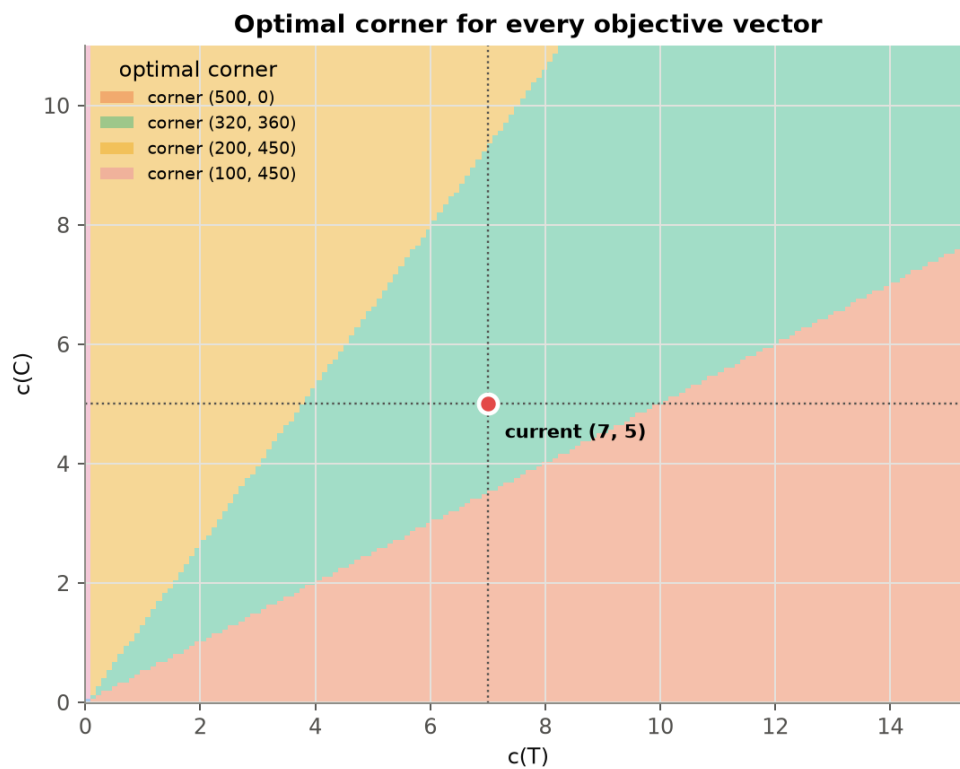
- **T = 320.** Its objective coefficient (7) may vary in [3.75, 10] without changing the optimal solution (one change at a time).
- **C = 360.** Its objective coefficient (5) may vary in [3.5, 9.333] without changing the optimal solution (one change at a time).
- **Carpentry** is binding. Shadow price 0.6: each extra unit of RHS changes Profit (\$) by 0.6, valid while the RHS stays in [1,500, 2,625].
- **Painting** is binding. Shadow price 2.6: each extra unit of RHS changes Profit (\$) by 2.6, valid while the RHS stays in [850, 1,600].
- **MaxChairs** is not binding (slack 90), so its shadow price is 0; the RHS can move within [360,  $\infty$ ] without effect on Profit (\$).
- **MinTables** is not binding (slack 220), so its shadow price is 0; the RHS can move within [ $-\infty$ , 320] without effect on Profit (\$).

## Sensitivity dashboard

Tables and Chairs: sensitivity dashboard ( $z^* = 4,040$ )

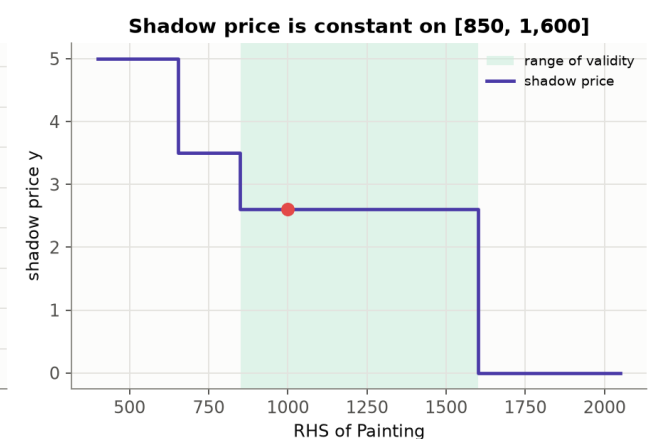
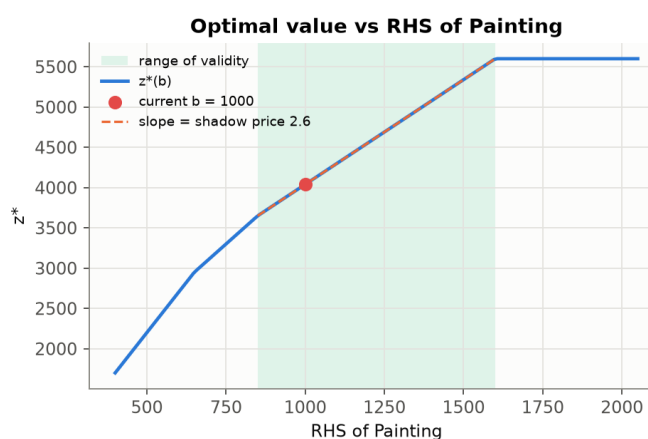
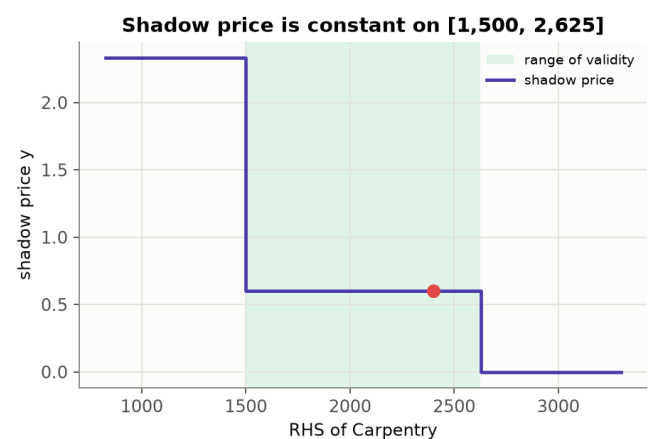
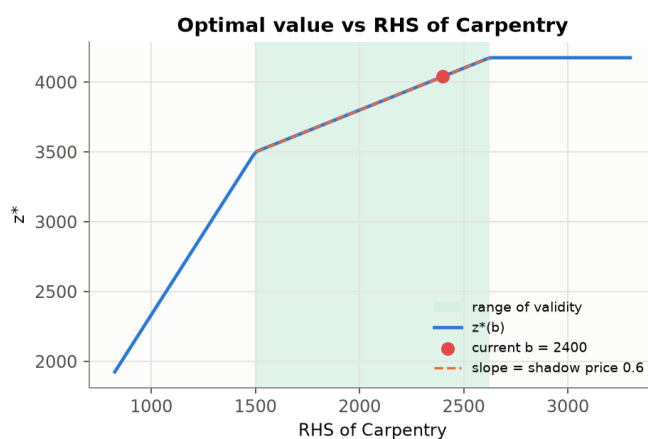
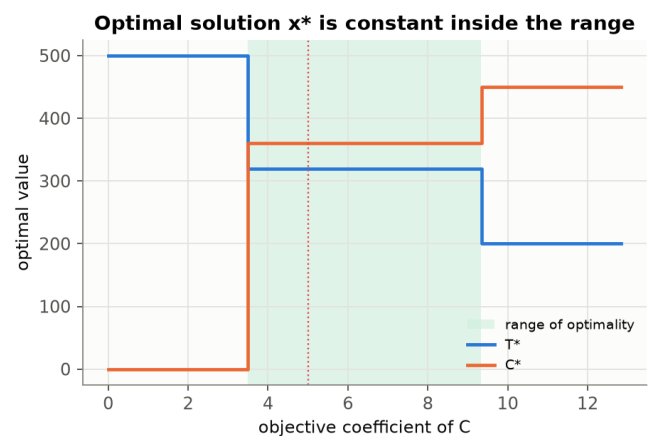
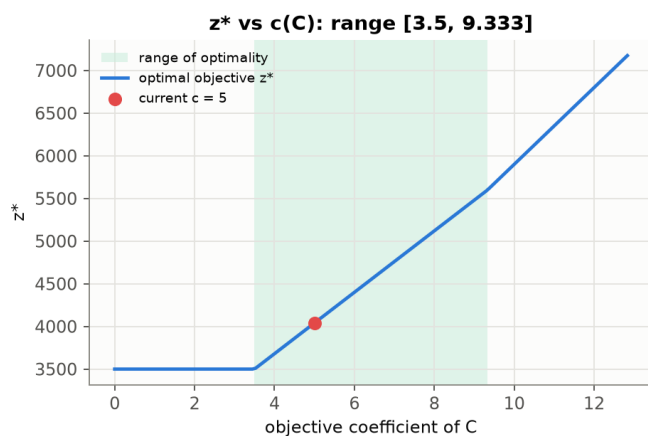
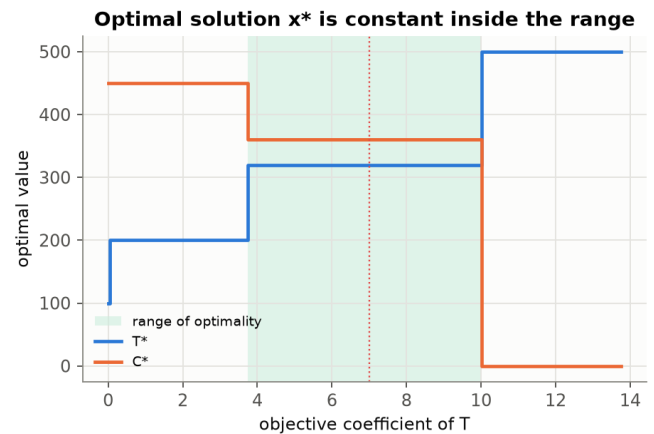
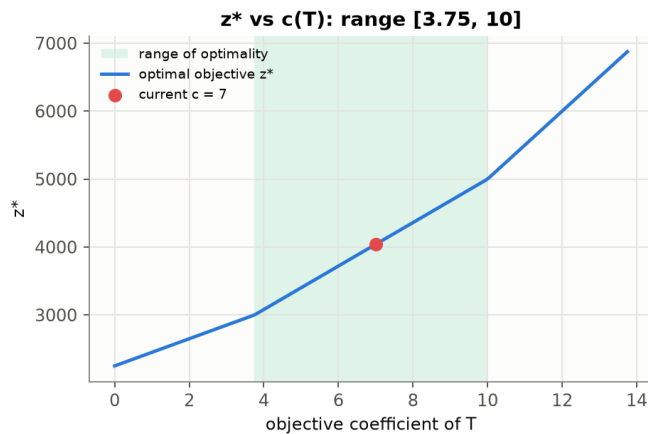


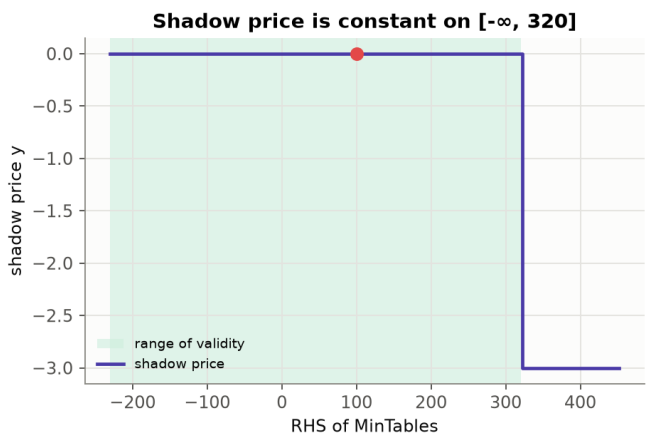
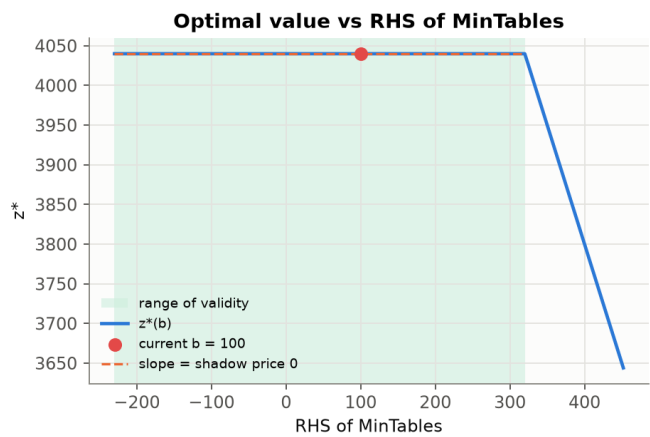
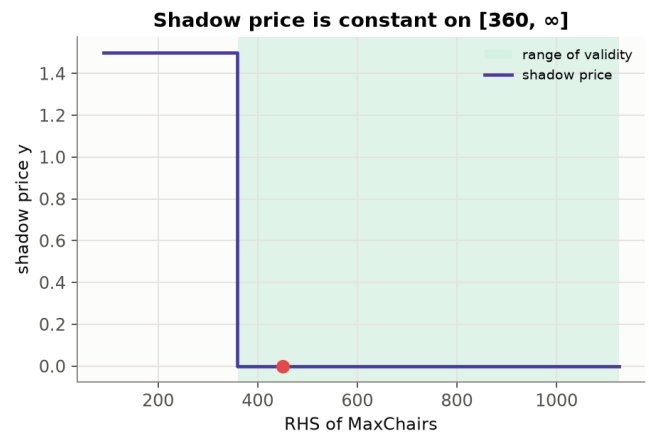
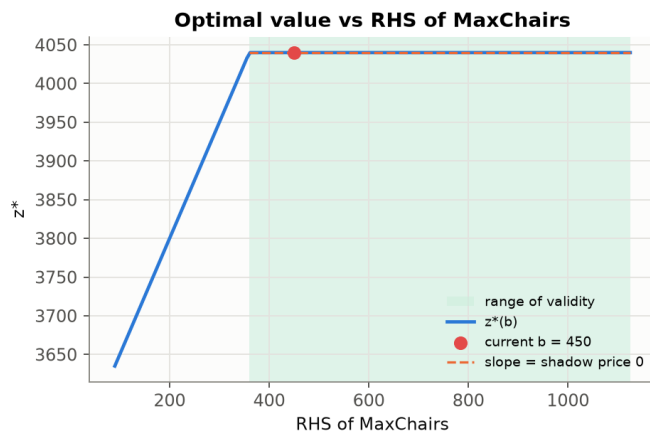
## Optimal corner for every objective vector



## 4. Parametric analysis

Each chart re-solves the LP for a range of values of one coefficient. The shaded band is the range reported in section 3.



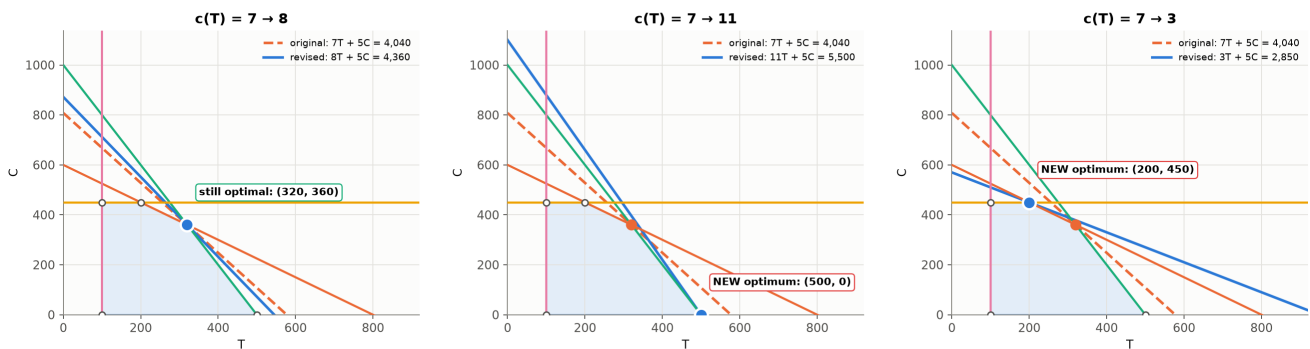


## 5. What-if scenarios

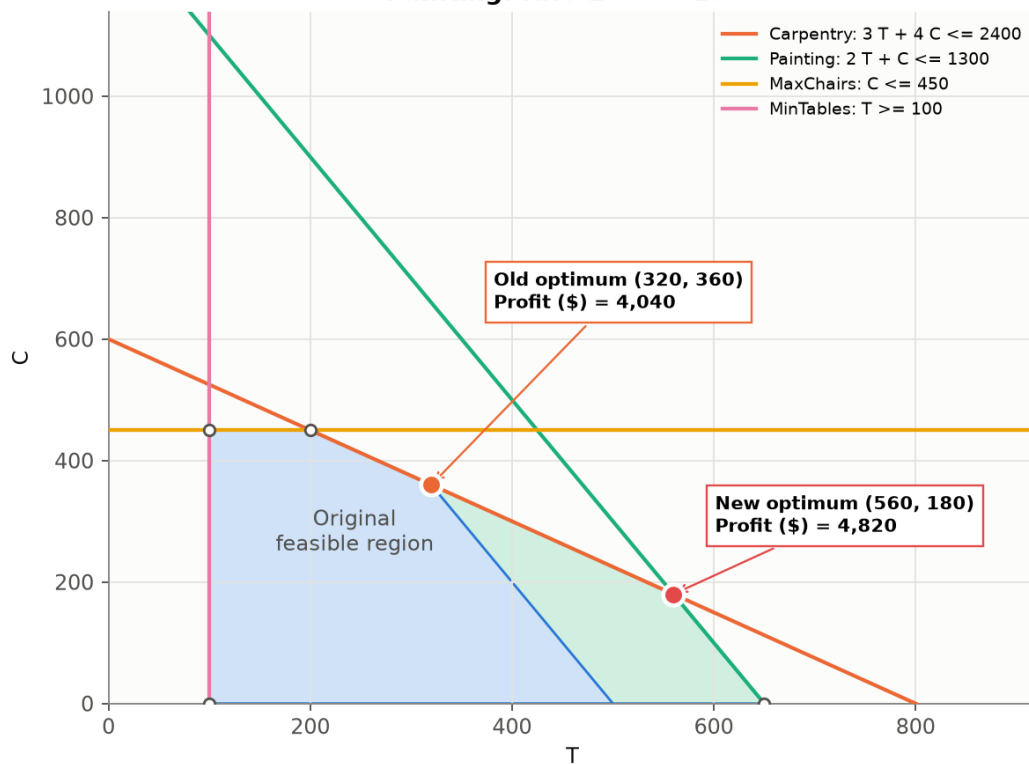
'Predicted' uses only the sensitivity report (shown when the change is inside its range); 'Actual' re-solves the LP.

| Scenario                 | InRange | Predicted | Actual | NewSolution | Note                                     |
|--------------------------|---------|-----------|--------|-------------|------------------------------------------|
| c(T): 7 → 8              | yes     | 4,360     | 4,360  | 320, 360    | slide 8/10: profit per table 7 -> 8      |
| c(T): 7 → 11             | no      | –         | 5,500  | 500, 0      | slide 11: profit per table 11            |
| c(T): 7 → 3              | no      | –         | 2,850  | 200, 450    | slide 11: profit per table 3             |
| b(Painting): 1000 → 1300 | yes     | 4,820     | 4,820  | 560, 180    | slide 14/16: painting hours 1000 -> 1300 |

Changing an objective coefficient rotates the iso-profit line; the feasible region never moves



Painting: RHS 1000 → 1300



## 6. Duality

### Dual formulation

MIN  $z = 2400 y_{\text{Carpentry}} + 1000 y_{\text{Painting}} + 450 y_{\text{MaxChairs}} + 100 y_{\text{MinTables}}$   
 subject to  
 $3 y_{\text{Carpentry}} + 2 y_{\text{Painting}} + y_{\text{MinTables}} \geq 7 \quad (\text{T})$   
 $4 y_{\text{Carpentry}} + y_{\text{Painting}} + y_{\text{MaxChairs}} \geq 5 \quad (\text{C})$   
 $y_{\text{Carpentry}} \geq 0, y_{\text{Painting}} \geq 0, y_{\text{MaxChairs}} \geq 0, y_{\text{MinTables}} \leq 0$

### Dual solution (resource / requirement prices)

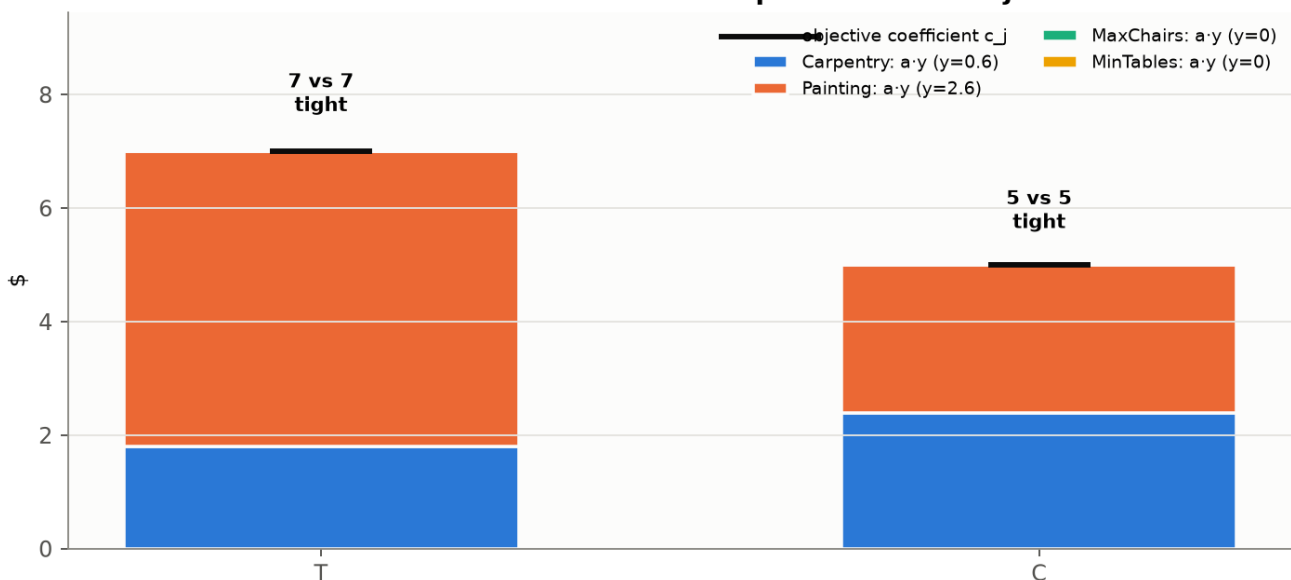
| Dual variable          | Primal constraint | Value $y^*$ | Primal shadow price | $b_i \cdot y_i$ |
|------------------------|-------------------|-------------|---------------------|-----------------|
| $y_{\text{Carpentry}}$ | Carpentry         | 0.6         | 0.6                 | 1,440           |
| $y_{\text{Painting}}$  | Painting          | 2.6         | 2.6                 | 2,600           |
| $y_{\text{MaxChairs}}$ | MaxChairs         | 0           | 0                   | 0               |
| $y_{\text{MinTables}}$ | MinTables         | 0           | 0                   | 0               |

Strong duality check: primal  $z^* = 4,040$ , dual  $w^* = 4,040$  (equal ✓). Dual of the dual equals the primal: yes.

### Complementary slackness (all products must be 0)

| Pair                                          | Slack | DualValue | Product |
|-----------------------------------------------|-------|-----------|---------|
| Carpentry slack $\times y_{\text{Carpentry}}$ | 0     | 0.6       | 0       |
| Painting slack $\times y_{\text{Painting}}$   | 0     | 2.6       | 0       |
| MaxChairs slack $\times y_{\text{MaxChairs}}$ | 90    | 0         | 0       |
| MinTables slack $\times y_{\text{MinTables}}$ | 220   | 0         | 0       |
| $T \times$ surplus of dual constraint T       | 0     | 320       | 0       |
| $C \times$ surplus of dual constraint C       | 0     | 360       | 0       |

**Dual constraints: value of resources in each product vs its objective coefficient**

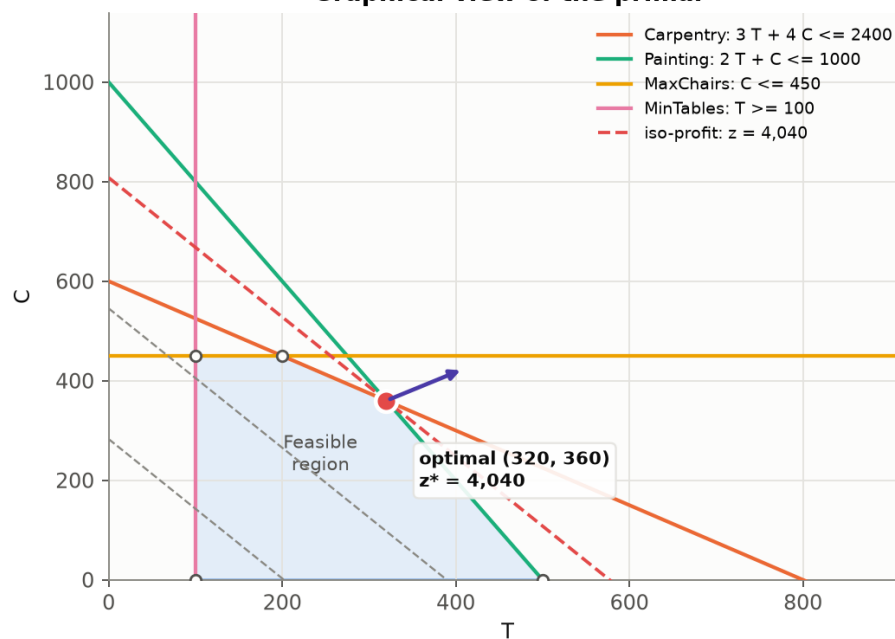


**Primal (max): 4 constraints × 2 variables**

| 7 | 5 | ← objective |
|---|---|-------------|
| 3 | 4 | ≤ 2400      |
| 2 | 1 | ≤ 1000      |
| 0 | 1 | ≤ 450       |
| 1 | 0 | ≥ 100       |
| T | C |             |

**Dual (min): 2 constraints × 4 variables**

| 2400        | 1000       | 450         | 100         | ← objective |
|-------------|------------|-------------|-------------|-------------|
| 3           | 2          | 0           | 1           | ≥ 7         |
| 4           | 1          | 1           | 0           | ≥ 5         |
| y_Carpentry | y_Painting | y_MaxChairs | y_MinTables |             |

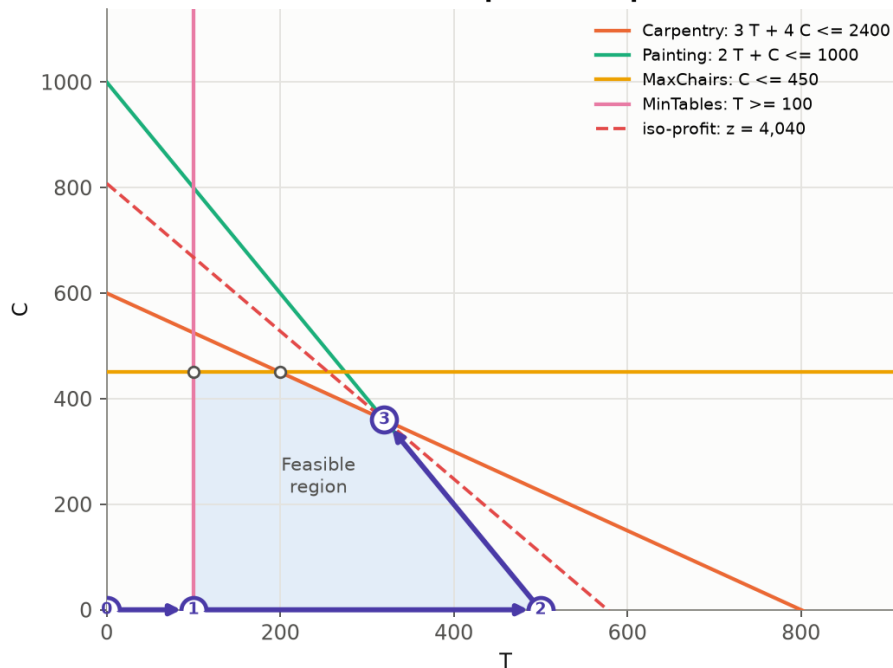
**Graphical view of the primal**

## 7. Simplex iterations

Two-phase tableau simplex (entering column = most negative  $z_i - c_i$ ; leaving row = minimum ratio).

| Tableau | Method | Phase | Entering | Leaving | Objective | PrimalFeasible | DualFeasible | Point      |
|---------|--------|-------|----------|---------|-----------|----------------|--------------|------------|
| 0       | primal | 1     | T        | a4      | 100       | yes            | no           | (0, 0)     |
| 1       | primal | 1     |          |         | -0        | yes            | yes          | (100, 0)   |
| 2       | primal | 2     | e4       | s2      | 700       | yes            | no           | (100, 0)   |
| 3       | primal | 2     | C        | s1      | 3,500     | yes            | no           | (500, 0)   |
| 4       | primal | 2     |          |         | 4,040     | yes            | yes          | (320, 360) |

Path of the primal simplex



Primal simplex - phase 1, tableau 0  
enter T, leave a4 (min ratio 100)

|    | T  | C | s1 | s2 | s3 | e4 | a4 | RHS  |
|----|----|---|----|----|----|----|----|------|
| w  | -1 | 0 | 0  | 0  | 0  | 1  | 0  | -100 |
| s1 | 3  | 4 | 1  | 0  | 0  | 0  | 0  | 2400 |
| s2 | 2  | 1 | 0  | 1  | 0  | 0  | 0  | 1000 |
| s3 | 0  | 1 | 0  | 0  | 1  | 0  | 0  | 450  |
| a4 | 1  | 0 | 0  | 0  | 0  | -1 | 1  | 100  |

Primal simplex - phase 1, tableau 1  
end of phase 1

|    | T | C | s1 | s2 | s3 | e4 | a4 | RHS  |
|----|---|---|----|----|----|----|----|------|
| w  | 0 | 0 | 0  | 0  | 0  | 0  | 1  | 0    |
| s1 | 0 | 4 | 1  | 0  | 0  | 3  | -3 | 2100 |
| s2 | 0 | 1 | 0  | 1  | 0  | 2  | -2 | 800  |
| s3 | 0 | 1 | 0  | 0  | 1  | 0  | 0  | 450  |
| T  | 1 | 0 | 0  | 0  | 0  | -1 | 1  | 100  |

**Primal simplex - phase 2, tableau 2**  
enter e4, leave s2 (min ratio 400)

|    | T | C  | s1 | s2 | s3 | e4 | RHS  |
|----|---|----|----|----|----|----|------|
| z  | 0 | -5 | 0  | 0  | 0  | -7 | 700  |
| s1 | 0 | 4  | 1  | 0  | 0  | 3  | 2100 |
| s2 | 0 | 1  | 0  | 1  | 0  | 2  | 800  |
| s3 | 0 | 1  | 0  | 0  | 1  | 0  | 450  |
| T  | 1 | 0  | 0  | 0  | 0  | -1 | 100  |

**Primal simplex - phase 2, tableau 3**  
enter C, leave s1 (min ratio 360)

|    | T | C    | s1 | s2   | s3 | e4 | RHS  |
|----|---|------|----|------|----|----|------|
| z  | 0 | -1.5 | 0  | 3.5  | 0  | 0  | 3500 |
| s1 | 0 | 2.5  | 1  | -1.5 | 0  | 0  | 900  |
| e4 | 0 | 0.5  | 0  | 0.5  | 0  | 1  | 400  |
| s3 | 0 | 1    | 0  | 0    | 1  | 0  | 450  |
| T  | 1 | 0.5  | 0  | 0.5  | 0  | 0  | 500  |

**Primal simplex - phase 2, tableau 4**  
optimal tableau

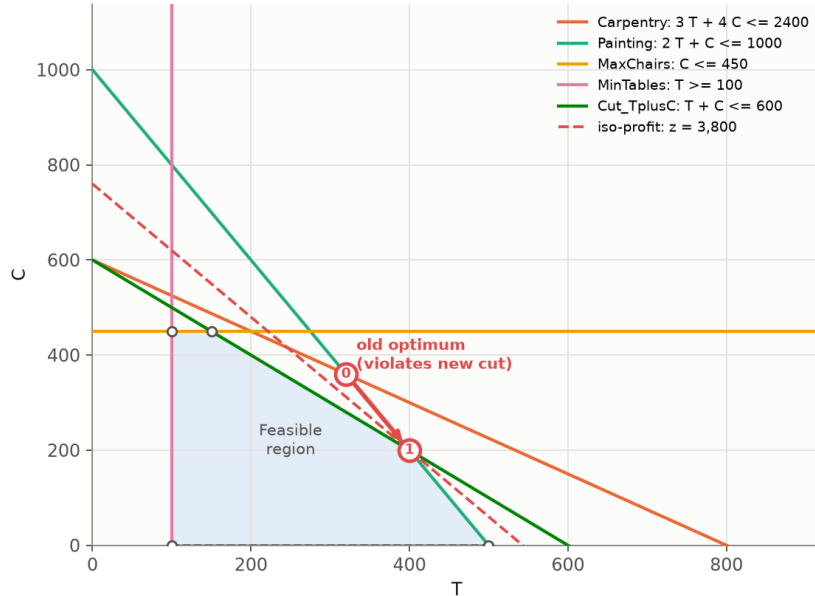
|    | T | C | s1   | s2   | s3 | e4 | RHS  |
|----|---|---|------|------|----|----|------|
| z  | 0 | 0 | 0.6  | 2.6  | 0  | 0  | 4040 |
| C  | 0 | 1 | 0.4  | -0.6 | 0  | 0  | 360  |
| e4 | 0 | 0 | -0.2 | 0.8  | 0  | 1  | 220  |
| s3 | 0 | 0 | -0.4 | 0.6  | 1  | 0  | 90   |
| T  | 1 | 0 | -0.2 | 0.8  | 0  | 0  | 320  |

## Dual simplex after adding 'Cut\_TplusC'

New constraint:  $1 T + 1 C \leq 600$ . dual simplex demo: combined capacity  $T + C \leq 600$  Result: **optimal**, new optimum (400, 200), Profit (\$) = 3,800.

| Tableau | Method | Phase | Entering | Leaving      | Objective | PrimalFeasible | DualFeasible | Point      |
|---------|--------|-------|----------|--------------|-----------|----------------|--------------|------------|
| 0       | dual   | 2     | s1       | s_Cut_TplusC | 4,040     | no             | yes          | (320, 360) |
| 1       | dual   | 2     |          |              | 3,800     | yes            | yes          | (400, 200) |

**Dual simplex: start outside the new region (primal infeasible), end at its optimum**



**Dual simplex - phase 2, tableau 0**  
leave s\_Cut\_TplusC (RHS -80 < 0), enter s1 (min |ratio| 3)

|              | T | C | s1   | s2   | s3 | e4 | s_Cut_TplusC | RHS  |
|--------------|---|---|------|------|----|----|--------------|------|
| z            | 0 | 0 | 0.6  | 2.6  | 0  | 0  | 0            | 4040 |
| C            | 0 | 1 | 0.4  | -0.6 | 0  | 0  | 0            | 360  |
| e4           | 0 | 0 | -0.2 | 0.8  | 0  | 1  | 0            | 220  |
| s3           | 0 | 0 | -0.4 | 0.6  | 1  | 0  | 0            | 90   |
| T            | 1 | 0 | -0.2 | 0.8  | 0  | 0  | 0            | 320  |
| s_Cut_TplusC | 0 | 0 | -0.2 | -0.2 | 0  | 0  | 1            | -80  |

**Dual simplex - phase 2, tableau 1**  
optimal (primal feasible again)

|    | T | C | s1 | s2 | s3 | e4 | s_Cut_TplusC | RHS  |
|----|---|---|----|----|----|----|--------------|------|
| z  | 0 | 0 | 0  | 2  | 0  | 0  | 3            | 3800 |
| C  | 0 | 1 | 0  | -1 | 0  | 0  | 2            | 200  |
| e4 | 0 | 0 | 0  | 1  | 0  | 1  | -1           | 300  |
| s3 | 0 | 0 | 0  | 1  | 1  | 0  | -2           | 250  |
| T  | 1 | 0 | 0  | 1  | 0  | 0  | -1           | 400  |
| s1 | 0 | 0 | 1  | 1  | 0  | 0  | -5           | 400  |